User preparation confirmation
用户准备确认单

Dear customer:

尊敬的客户：

Thank you for choosing our company's products. In order to ensure the smooth on-site installation and commissioning of the equipment and to deliver it for use as soon as possible, the following preparations shall be completed in advance before the equipment installation. Please confirm according to the following table.

感谢您选择我公司产品，为保证本设备顺利进行现场安装调试，尽快交付使用，设备安装前需预先完成以下准备工作，请按照下方表格进行确认。

After the following work is completed, please sign and seal this confirmation form and send it back to our company, so that we can provide you with equipment installation services on time.

以下工作内容完成以后，请将本确认表签字盖章后回传至我公司，以便我方按时为贵方提供设备安装服务。

If the installation time is extended due to insufficient preparation, we will charge the service fee according to the *BODOR Fiber Laser Machine Warranty Regulation*.

如果因准备不足而导致的安装时间延长，我们将按照《邦德激光切割机保修规定》收取服务费用。

Jinan Bodor CNC Machine Co., Ltd.
济南邦德激光股份有限公司

Requirements for site installation conditions 现场安装条件要求：

Condition requirement 条件要求	Required scope 要求范围	Remarks 备注	confirm 确认“√”
Site requirements 场地要求	The installation foundation of the equipment shall be flat and the flatness shall be within ± 5mm, and the thickness of the concrete on the whole installation surface shall not be less than 150mm(H,G,L,DREAM,M,U≥200mm). 设备的安装地基要求平整，平整度应该在±5mm 以内，整个安装面的混凝土的厚度不低于 150mm （H,G,L,DREAM,M,U ≥ 200mm）。	If the requirements cannot be met, normal installation of the equipment cannot be guaranteed. 如不能达到要求无法保证设备正常安装。	
Foundation 地基	The installation site must make the foundation according to the foundation drawing of our company. 安装场地必须按照我公司地基图纸制作地基。	If the foundation is not made according to the requirements, the equipment installation cannot be carried out. 如未按照要求制作地基，无法进行设备安装。	
Space requirement 空间要求	Arrange the site according to the site layout provided by our company. In order to prevent the occurrence of fire, the processing site should be equipped with appropriate fire extinguishers and reserve a certain fire extinguishing channel. 按照我公司提供的现场布局图布置场地。为预防火灾的发生，加工场地应配置好适当的灭火器并预留一定的灭火通道。	If the space is too small, it will cause inconvenience in the use and maintenance of equipment in the future, and there will be certain hidden dangers in the safety of users. 如空间过小，会造成今后的设备使用及维修不便，使用者的安全也会存在一定的安全隐患。	
Environmental requirements 环境要求	The equipment shall be kept in a dry, well ventilated environment with an ambient temperature between + 4 ℃ and + 33 ℃ and a humidity of less than 75%. When the equipment is shut down, the ambient temperature at the installation site shall not fall below + 4 ℃. The place shall be kept away from direct sunlight, and shall not have rain leakage or water seepage. 应保持设备的运行在干燥、通风良好、环境温度在 +4℃ 与 +33℃ 之间，湿度必须小于75%的环境下。设备关机状态下，安装地点的环境温度不得降到 +4℃以下。放置场地应避免阳光直射，不能有漏雨、渗水的情况发生。	Too low temperature will cause freezing damage of the equipment, and too high temperature will cause condensation. Areas that do not meet environmental requirements need to be equipped with air-conditioned rooms for laser source. 温度过低会造成设备的冻坏，过高会造成结露情况。不符合环境要求的区域需要为激光器配置空调房间。	
Environmental vibration 环境震动	There is no violent vibration source within 50m around the equipment. If there is, anti vibration ditch shall be arranged. 设备周边50米内无剧烈震动源，若有应布置防震沟。	Site vibration will cause equipment failure and affect cutting effect and accuracy. The layout of shockproof ditch shall meet the technical specifications for shockproof requirements. 场地震动会造成设备故障，影响切割效果和精度。防震沟布置应符合防震要求技术规范。	
Power quality 电源品质	The total capacity shall not be lower than the equipment requirements, the stability shall be less than 5%, and the three-phase imbalance shall be less than 2.5%. For areas where industrial power supply is AC380V, the customer's on-site incoming line requirements must have 3 live wires, 1 neutral wire, and 1 ground wire; For areas where industrial power supply is not AC380V, the incoming line at the customer's site must have 3 live wires and 1 ground wire, and there is no need to connect the neutral wire to the transformer. 总容量不低于设备要求，稳定度小于5%，三相不平衡小于2.5%。对于工业供电是AC380v的地区，客户现场的进线要求必须具备3条火线，1条中性线和1条地线；对于工业供电为非AC380v的地区，客户现场的进线要求必须具备3条火线和1条地线，不需要中性线接入变压器。	If the site does not meet the requirements, it must be equipped with a voltage regulator, otherwise it will cause equipment failure. 如现场不具备要求，必须配备稳压器，否则会造成设备故障。	
Environmental quality 环境质量	In order not to affect the cutting quality, it is necessary to ensure that there are no volatile substances or particles generated around the equipment, such as solvent containing steam emitted during painting, steam emitted from oil removal equipment, or local environmental humidity caused by water jet cutting. 为了不影响切割质量，设备周围必须确保没有挥发性物质或者颗粒产生，例如在喷漆时散发出的含溶剂的蒸汽、从除油设备上发出的蒸汽或者水刀切割引起局部环境湿度变大的情况。	Failure to meet the requirements may result in lens and optical fiber pollution, which may affect the cutting effect. 如达不到要求可能会照成镜片及光纤的污染，影响切割效果。	

Earthing protection 接地保护	The machine is grounded separately, and the grounding resistance is required to be less than 4 Ω. If the ground wire is not lead into the customer's site, need customer to make the ground wire on site. Ground wire requirements: use triangle iron or coarse steel or brass/copper rod 欧or galvanized rod longer than two meters long to smash into the ground, and pour proper amount of salt water at the same time to ensure good grounding effect, and then lead out a high-quality stranded soft copper wire with cross-sectional area no less than the main power cable cross-sectional area of the machine. 机床单独接地，要求接地电阻＜4Ω。如果供电端没有引入地线，则需要客户在现场制作地线。地线要求：用长于两米的三角铁、粗钢筋、铜棒或者镀锌棒砸入地下，同时灌注适量盐水，确保接地效果良好，然后引出一根优质的多股绞合软铜线，截面积不小于机床总电源线截面积。	In order to ensure the normal use and electrical safety of the equipment, it must be configured according to this requirement. It is forbidden to ground the ground wire on the steel column of the steel structure workshop. 为确保设备的正常使用及电气安全，必须按此要求配置。禁止在钢构厂房钢柱上接地线。	

Work required for installation 安装就位所需工作：

Name 名称	Requirement 要求	confirm 确认“√”
Equipment transportation 设备运输	Choose appropriate tonnage of cranes, truck cranes, forklifts, electric heavy-duty universal transport vehicles and other handling equipment according to the weight of the equipment, and consult customer service staff before unloading. 根据设备重量选择合适吨位的行车、汽车吊车、叉车、电动重型万向运输车等搬运设备，卸货前咨询客服人员。	
Power 电源	The power supply shall be equipped with independent main switch and shall not be connected with other equipment. 该电源应有独立总闸，不要和其他设备连接在一起。	
Power Distribution Box 配电箱	The distribution box is equipped with separate air switch for equipment and auxiliary equipment. 配电箱内装设设备及辅机单独空开。	
Power cable 电源线	For the equipment equipped with transformer or voltage regulator, please prepare the cable connecting the transformer or voltage regulator to the equipment according to the power consumption of the equipment and the cable length required on site. 对于配置了变压器或稳压器的设备，请根据设备用电功率及现场需要线缆长度准备变压器或者稳压器连接到设备的电缆。	
Electric welding machine 电焊机	1 set, a number of welding rods, used for fixing the foundation. 1台，焊条若干，用于地脚的固定。	
Percussion drill 冲击钻	Used for fixing equipment fences and equipment footings. 用于设备围栏和设备地脚的固定。	
Dust removal pipeline 除尘管道	Machines equipped with exhaust auxiliary devices will be delivered with some exhaust pipes of the machine. If the field fan is placed too far away and exceeds the length of its own pipe, it needs to be equipped by the customer. 对于配置排风外设的机器配有排风管道。如果现场风机放置过远，超出自带管道长度，需客户配备。	

Work required for commissioning 调试所需工作：

Name 名称	Requirement 要求	Remarks 备注	confirm 确认“√”
Test cutting plate 试切板材(板切机设备)	There are several 400mmx400mm plates, the thickness of which shall be prepared according to the cutting requirements on site. 400mmx400mm板材若干，板材材料厚度应根据现场切割需求准备。	Suggested carbon steel grade: Q235, stainless steel grade: 201, 304. 建议碳钢标号：Q235，不锈钢标号：201、304。 The plate surface shall be free of rust, paint and oil stain. Otherwise, the cutting effect will be affected. 板材表面无锈，无油漆、无油污。否则会影响切割效果。	
Test cutting tube 试切管材(管切机设备)	A number of pipes with a length of no less than 1.5m shall be prepared according to the cutting requirements on site. 长度不小于1.5m的管材若干，规格型号根据现场切割需求准备。	The pipe is invisible, and the surface is free of rust, paint and oil stain. 管材无形变，表面无锈，无油漆，无油污。	
Oxygen 氧气	Purity ≥ 99.9%, outlet pressure ≥ 1.0Mpa, gas quality standards comply with ISO 8573-1:2010 Solid Particles - Grade 2, Water - Grade 4, Oil - Grade 3. 纯度≥99.9%，出气压力≥1.0Mpa，气体质量标准符合ISO 8573-1:2010固体颗粒-等级 2，水-等级4，油-等级3。	Liquid oxygen is preferred, with a suitable vaporizer (≥ 25m ³ / h). 液氧优先，配合适汽化器（≥25m³/h）。	
Oxygen meter 氧气表	The customer needs to prepare an oxygen pressure reducing valve to connect the air source to the equipment. The specification of the oxygen reducing valve: the input pressure range is 25MPa, the output range is 2.5MPa, the air source connected at the inlet end meets the local standards, and the Pu gas pipe with an outer diameter of 10mm and an inner diameter of 6mm can be connected at the outlet end. 需要客户配置氧气减压阀一个，用于连接气源至设备，氧气表规格：输入压力量程25Mpa，输出量程2.5Mpa，进气端连接气源符合当地标准，出气端可以连接外径10mm内径6mm的PU气管。	Clamps are required to connect tower joints. 连接塔型接头需要配备卡箍。	

Nitrogen 氮气	Purity ≥ 99.99%, outlet pressure should be 2.5MPa, gas quality standards comply with ISO 8573-1:2010 Solid Particles - Grade 2, Water - Grade 4, Oil - Grade 3. 纯度≥99.99%，出气压力需达到2.5Mpa，气体质量标准符合ISO 8573-1:2010固体颗粒-等级 2，水-等级4，油-等级3。	Liquid nitrogen is preferred and needs to be equipped with vaporizer, laser power ≤ 4kw, vaporizer ≥ 150 m³/h; laser power ≥ 6kW, vaporizer ≥ 200 m³/h; laser power ≥ 10kW, vaporizer ≥ 300m m³/h 液氮优先，需要配汽化器，激光器功率≤4kw，汽化器≥150m³/h；激光器功率≥6kw，汽化器≥ 200m³/h；激光器功率≥10kw，汽化器≥ 300m³/h	
Nitrogen meter 氮气表	The customer needs to prepare a nitrogen pressure reducing valve to connect the air source to the equipment. The specification of nitrogen reducing valve: The plate cutting machine with the laser power ≤ 6kw （Excluding scan cutting machine and P-6kw machine）, all the tube cutting machine, the input pressure range is 25MPa, the output range is 6Mpa, the inlet end is connected with the gas source according to the local standard, and the outlet end can be connected with a nylon gas pipe with an outer diameter of 10mm and an inner diameter of 7.5mm. The laser power ≥ 12kW （Including scan cutting machine and P-6kw machine）, the input pressure range is 25MPa, the output range is 10MPa, the inlet end is connected with the gas source in accordance with the local standard, and the gas outlet end can be connected with the internal M22 * 1.5N gas pipe connector. 需要客户配置氮气减压阀一个，用于连接气源至设备，氮气表规格：板切机设备激光器功率≤6kw（扫描切和P-6kw设备除外），管切机全部设备，输入压力量程25Mpa，输出量程6Mpa，进气端连接气源符合当地标准，出气端可以连接外径10mm内径7.5mm的尼龙气管。激光器功率≥12kw（包含全部扫描切和P-6kw），输入压力量程25Mpa，输出量程10Mpa，进气端连接气源符合当地标准，出气端可以连接内丝M22*1.5N的气管接头。	Clamps are required to connect tower joints. 连接塔型接头需要配备卡箍。	
compressed air 压缩空气	Compressed air source (required for equipment function use) 压缩空气气源（设备功能使用，必配）	It is required to be clean, anhydrous, oil-free and other impurities, with air supply volume of 0.8 m³ / min and pressure no less than 0.6Mpa. 要求洁净、无水、无油等杂质，供气量：0.8 m³/min,压力不低于0.6Mpa。	
	Special air compressor for laser cutting (used as cutting auxiliary gas, optional). 激光切割专用空压机（作为切割辅助气体使用，选配）	It is required to be clean, Water-free, oil-free and no other. It is used for cutting carbon steel and stainless steel when conditions permit. (the key parameters of the air compressor shall not be lower than the following standards: please use the screw air compressor, whose volume flow shall meet the laser power, the maximum working pressure ≥ 1.6Mpa, the pressure dew point 2 ~ 8 ℃, and the filtering accuracy 0.01 μm. Oil content at gas outlet ≤ 0.003ppm, gas quality standards comply with ISO 8573-1:2010 Solid Particles - Grade 2, Water - Grade 4, Oil - Grade 3.) 要求洁净、无水、无油等杂质，用于条件允许的碳钢、不锈钢切割。（空压机关键参数不得低于以下标准：请使用螺杆式空压机，其容积流量需满足激光器功率，最大工作压力≥1.6mpa，压力露点2~8℃，过滤精度0.01μm，出气口含油量≤0.003ppm，气体质量标准符合ISO 8573-1:2010固体颗粒-等级 2，水-等级4，油-等级3。）	
Distilled or purified water 蒸馏水或纯净水	High quality mineral free purified water, distilled water or deionized water is required. 要求使用高质量的不含矿物质的纯净水、蒸馏水或者去离子水。	The specific demand volume varies with the power, please consult the customer service engineer. 具体需求体积因功率不同而需求不同，请咨询客服人员。	
Hydraulic oil 液压油	46# Hydraulic oil 46#液压油	Due to cross-border transportation requirements, the hydraulic oil in equipment with hydraulic pumps needs to be drained, so customers need to purchase and supplement it themselves. Most of our equipment do not have hydraulic pumps. Please consult customer service engineer for specific preparation. 因跨国运输要求，带有液压泵的设备内的液压油需要排空，故需要客户自行购买补充，我司绝大多数设备都没有液压泵，具体是否需要准备请咨询客服人员。	
Wide body transparent tape 宽体透明胶带	1 volume 1卷	Adjust the laser focus coaxial。 调整激光焦点同轴。	
Computer 电脑	1 set 1台	The system is the flagship version of Windows 10 and 11, and the computer has d-disk partition. 系统为Windows10/11旗舰版，且电脑有D盘分区。	
Smart mobile phone/tablet computer 手机/平板电脑	1set, Connection and use of intelligent monitoring system for laser heads of 12kw and above 1set，12kw及以上激光头智能监控系统连接使用	The system is Android 11 or above, with Bluetooth transmission function, and can install APK software packages. 系统为安卓11及以上版本，具备蓝牙传输功能，可以安装APK软件包。	

Staffing 人员配备：

2-4 programming operators are required on site (preferably with CAD drawing experience and CNC equipment operation experience). In the process of equipment installation and debugging, there should be the participation and assistance of programming operators, so that they can learn in the installation process, which will be very helpful to familiarize and operate the equipment.

现场需配备2-4名编程操作人员（有CAD制图经验和数控设备操作经验人员为佳）。设备安装调试过程中要有编程操作人员的参与和协助，这样可以在安装过程中进行学习，对于熟悉和操作设备会有很大的帮助。

Key points 重点提示：

The above conditions must be fully met before our service personnel can carry out the installation and commissioning of the equipment. If the customer fails to complete the above preparatory work as required and insists on our personnel to carry out the installation and commissioning of the equipment, a pre installation confirmation sheet must be signed, indicating that the resulting extension of the installation and commissioning lead time, the increase of service costs and the equipment failure, performance degradation, stability and efficiency reduction caused by the failure of the installation conditions in the future use of the equipment The consequences shall be borne by the customer, and bond laser shall not bear any liability and joint expenses arising therefrom.

以上条件需完全具备后我方服务人员才可进行设备的安装调试。如客户未按照要求完成以上准备工作，坚持让我方人员进行设备的安装调试，必须签署设备安装前确认单，表明由此产生的安装调试交付周期延长、服务费用增加以及在今后设备使用中因安装条件达不到要求而导致的设备故障、性能下降、稳定性及效率降低等不良后果，均由客户自行承担，邦德激光不承担由此产生的任何责任和连带费用。

Confirmation sheet before equipment installation
设备安装前确认单

Incomplete site preparation: 现场准备工作未完成项:	
Reasons for incomplete site preparation: 现场准备工作未完成原因	☐Bodor reasons邦德原因 ☐Customer reasons客户原因 ☐Reason to be determined原因待定: reason原因简述_____ _____ _____ _____ _____

Note: if the customer site did not meet the installation and commissioning conditions of the equipment, the customer also did not follow the requirements of the customer site confirmation form to improve the site and material preparation. The resulting extension of the installation commissioning delivery cycle, increased service costs and the adverse consequences of equipment failure, performance degradation, stability and efficiency reduction caused by failure to meet the requirements in the future use of the equipment shall be borne by the customer, and Bond Laser shall not bear any liability and associated costs arising therefrom.

注意：如客户现场未达到设备的安装调试条件，客户也未按照客户现场确认单中的要求对现场进行完善和物料准备。由此产生的安装调试交付周期延长、服务费用增加以及在今后设备使用中因安装条件达不到要求而导致的设备故障、性能下降、稳定性及效率降低等不良后果，均由客户自行承担，邦德激光不承担由此产生的任何责任和连带费用。

Information confirmation
信息确认

Equipment model 设备型号	Order NO. 设备单号	Country 所在国家	Customer email 客户邮箱:
Customer company name 客户公司名称		Customer signature 客户签章	
		Date:	